



E.H

KNOX GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT

2001
TRIAL HSC EXAMINATION

Mathematics

Extension 2

- **General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided on page 10
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1–8
- All questions are of equal value
- Use a SEPARATE writing booklet for each question

NAME: _____

TEACHER: _____

Total marks (120)
Attempt questions 1 – 8
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) Find:

(i) $\int \frac{x}{\sqrt{9-4x^2}} dx$ **2**

(ii) $\int \frac{x^2}{x+1} dx$ **2**

(iii) $\int_0^{\ln 2} xe^x dx$ **3**

(b) (i) Find real numbers A , B and C such that $\frac{2}{(t+1)(t^2+1)} = \frac{A}{t+1} + \frac{Bt+C}{t^2+1}$. **3**

(ii) Hence, find $\int_0^1 \frac{2}{(t+1)(t^2+1)} dt$. **3**

(iii) By using the substitution $t = \tan\left(\frac{x}{2}\right)$ evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x - \cos x} dx$. **2**

Question 2 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) Suppose $z = 2 + 2i$ and $w = -1 + \sqrt{3}i$.

(i) Express z and w in modulus – argument form. **2**

(ii) Find $\left| \frac{z}{w} \right|^4$. **1**

(iii) Find the principal argument of $\left(\frac{z}{w} \right)^4$. **2**

(b) Sketch separately the following loci in an Argand plane and state the cartesian equations in each case given that:

(i) $|z - 3i| = |z - 4|$ **2**

(ii) $\operatorname{Re}\left(\frac{z-2}{2}\right) = 0$ **2**

(iii) $\arg(z+2) = -\frac{\pi}{6}$ **2**

(c) (i) Show that if $z = x + iy$ then $|z|^2 = z\bar{z}$. **1**

(ii) Using the result of (c)(i), or otherwise, prove that for any two complex numbers z and w that:

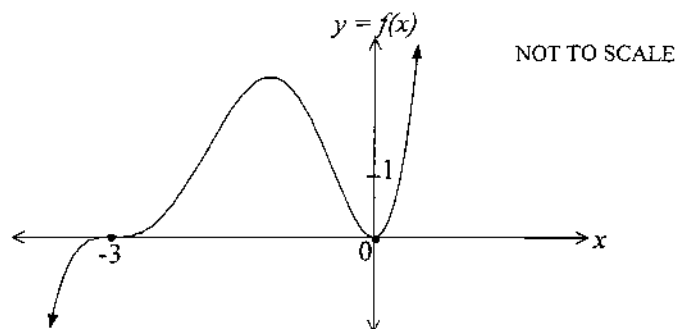
$$|z+w|^2 + |z-w|^2 = 2|z|^2 + 2|w|^2.$$
 2

(iii) Interpret this result geometrically. A vector diagram may be useful. **1**

Question 3 (15 marks)

Use a SEPARATE writing booklet

Marks



- (a) Consider the graph of $y = f(x)$ as shown above.

On the answer sheet provided on pages 11 & 12, use the graph of $y = f(x)$ to clearly sketch separately the graphs of:

(i) $y = \frac{1}{f(x)}$

(ii) $y^2 = f(x)$

(iii) $y = f'(x)$.

- (b) Suggest a possible polynomial equation for the graph of $y = f(x)$ shown in **part (a)** of **Question 3**.

- (c) (i) Show that $x = 1$ is a zero of $x^3 + 3x^2 - 4$.

- (ii) Sketch the curve with the equation $y = x^3 + 3x^2 - 4$, giving the coordinates of any maximum or minimum points and the intercepts made on each axis.

- (iii) Use your results in (c)(ii) above to sketch the curves:

(α) $y = |x^3 + 3x^2 - 4|$

(β) $y = \ln|x^3 + 3x^2 - 4|$

- (iv) Hence, or otherwise, determine the value of m , where m is a constant such that the equation $2 \ln|x+2| + \ln|x-1| = m$.

Question 4 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) (i) Draw a sketch graph of the hyperbola $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ and shade clearly the region bounded by the lines $x = \pm a$ and the upper and lower branches of this hyperbola.

- (ii) Show $\frac{d}{d\theta} \ln(\sec \theta + \tan \theta) = \sec \theta$.

- (iii) Explain why the area, A , of the shaded region drawn in (a)(i) above can be by:

$$A = \int_0^a \frac{4b}{a} \sqrt{a^2 + x^2} \, dx.$$

- (iv) By using the substitution $x = a \tan \theta$ in (a)(iii), show that $A = 4ab \int_0^{\frac{\pi}{4}} \sec^3 \theta \, d\theta$.

- (v) Show that the integral stated in (a)(iv) simplifies to $2ab(\sqrt{2} + \ln(\sqrt{2} + 1))$.

(**Hint:** Write $\sec^3 \theta$ in the form $\sec \theta \cdot \sec^2 \theta$ and then use integration by parts)

- (vi) Use the **method of cylindrical shells** to show that the volume (in cubic units) of the solid generated by revolving this area about the y -axis is given by:

$$V = \frac{4\pi ba^2}{3} (2\sqrt{2} - 1).$$

- (b) A solid has a base, which is the *standard ellipse* $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with major axis of length $2a$ units and minor axis of length $2b$ units ($a > b$). In the vertical plane, the cross-sections of the solid are always isosceles triangles with perpendicular height h and whose base is parallel to the major axis.

Use the **method of slicing** to find the volume of the solid.

Question 5 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) The point T with coordinates $(at^2, 2at)$, $t \neq 0$, $a > 0$, lies on the parabola with equation $y^2 = 4ax$. The tangent to the parabola at T meets the axis of the parabola at R . The normal at T meets the axis of the parabola at Q and the parabola again at P . The coordinates of P are $(ap^2, 2ap)$.

- (i) Represent this information on a clear and well-labelled diagram. 1
- (ii) Derive the equations of the tangent and normal to the parabola at T . 2
- (iii) Show that the length of RQ is $2a(1+t^2)$ units. 1
- (iv) Show that the values of t for which R will lie on the directrix of the parabola satisfy $t^2 = 1$. 2
- (v) Show that if $t \neq p$, then $p = -\left(t + \frac{2}{t}\right)$. 2
- (vi) Find TP , in terms of a and simplify your expression as far as possible. 1
- (vii) Hence, or otherwise, prove that the area of ΔTPR is $16a^2$ square units. 1
(You may assume R lies on the directrix)

- (b) The equation of a rectangular hyperbola in cartesian form is given by $xy = c^2$ where $c > 0$.

- (i) Verify that the point $P\left(cp, \frac{c}{p}\right)$ lies on $xy = c^2$, where p is a non-zero real number. 1
- (ii) Q has coordinates $\left(cq, \frac{c}{q}\right)$ where q is a non-zero real number. 2
Show that the equation of the chord PQ is given by $x + pqy = c(p + q)$.
- (iii) Find the equation of the locus of the midpoint of the chord PQ if it is known that the chord must always pass through the point $(0, 2)$. 2

Question 6 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) A particle of mass m units is projected vertically upward from the ground with initial speed u . The air resistance at any instance is proportional to the velocity v at that instant. For this question you may assume $R = kmv$ where k is a constant.

- (i) With the aid of a suitable diagram show that $\frac{dv}{dt} = -(g + kv)$? 1

- (ii) Show at any time t , that $t = \frac{1}{k} \ln \left| \frac{g + ku}{g + kv} \right|$ seconds. 3

- (iii) Prove that the particle reaches its highest point in time T seconds when: 1

$$T = \frac{1}{k} \ln \left(\frac{ku}{g} + 1 \right)$$

- (iv) The highest point reached by the particle is at H metres above the ground.

- (α) Prove that $x = \frac{1}{k^2} (g + ku) (1 - e^{-kt}) - \frac{gt}{k}$. 3

- (β) Prove that $H = \frac{1}{k} (u - gT)$. 2

- (b) Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n \theta \, d\theta$ where n is a positive integer such that $n \geq 2$.

- (i) By replacing $\sin^n \theta$ with $\sin^{n-1} \theta \cdot \sin \theta$, and using integration by parts or otherwise, show that $I_n = \frac{n-1}{n} I_{n-2}$. 3

- (ii) Hence, or otherwise, evaluate I_{10} . 2

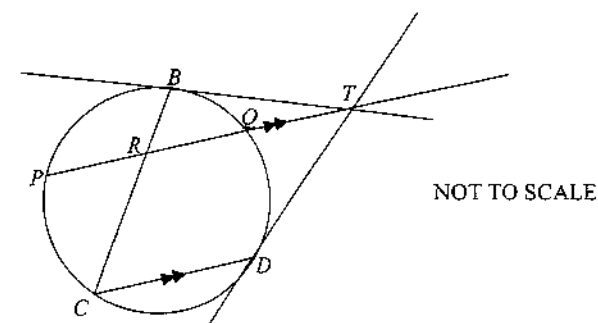
Question 7 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Let α, β , and γ be the non-zero roots of the equation $x^3 + rx + s = 0$.
- (i) Find in terms of r , the simplified value of $\alpha^2 + \beta^2 + \gamma^2$. 2
- (ii) Find in terms of r and s , the simplified value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$. 2
- (iii) Find in terms of r and s , the cubic equations (in general form) whose roots are
- (A) $\frac{1}{\alpha}, \frac{1}{\beta}$, and $\frac{1}{\gamma}$; 2
- (B) $\alpha + \beta - \gamma, \beta + \gamma - \alpha$ and $\gamma + \alpha - \beta$ 3
- (b) Suppose $x^3 + rx + s = 0$ (with r and s being non-zero and real) has a double root. 2
- Show that $x = -\frac{3s}{2r}$.
- (c) Find all the roots of $p(x) = x^4 - 8x^3 + 39x^2 - 122x + 170$ given that $3 - i$ is one of the roots. 4

Question 8 (15 marks) Use a SEPARATE writing booklet

Marks



- (a) In the diagram, PQ and CD are parallel chords of a circle. The tangent at D meets PQ produced externally at T . B is the point of contact of the other tangent from the circle. BC meets PQ internally at R .

Copy or trace this diagram into your writing booklet

- (i) Explain why $\angle BDT = \angle BRT$? 2
- (ii) Show that B, T, D and R are concyclic points. 2
- (iii) Prove that $\angle BRT = \angle DRT$. 2
- (iv) Show that $\triangle RCD$ is isosceles. 2
- (v) Show that BC bisects PQ . 2
- (b) (i) Show that $\cos x = \sin\left(x + \frac{\pi}{2}\right)$. 1
- (ii) Given that $y = 3 \sin x + 4 \cos x$, prove by the Principle of Mathematical Induction that $\frac{d^n y}{dx^n} = 5 \sin\left(x + \alpha + \frac{n\pi}{2}\right)$ where $\frac{d^n y}{dx^n}$ means the n th derivative of y with respect to x and $n = 1, 2, 3, \dots$. 4
- You are advised to first express $y = 3 \sin x + 4 \cos x$ in the form $R \sin(x + \alpha)$.

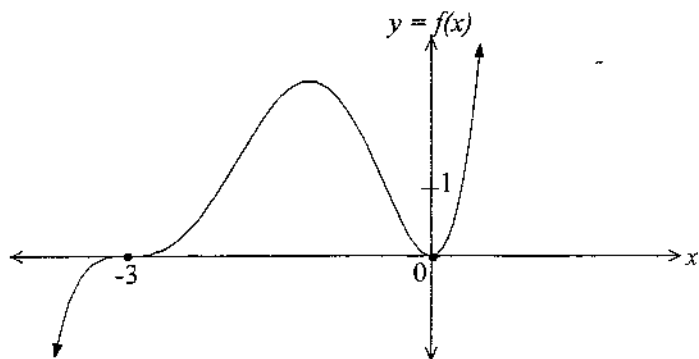
End of Paper

Detach and submit this page with your solutions to Question 3

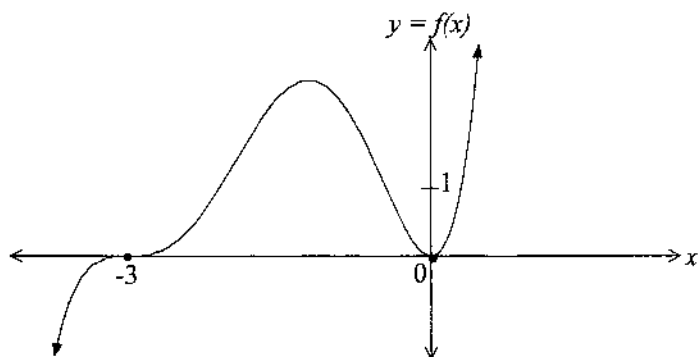
Student Name: _____

Question 3 (a) In each case use the graph of $y = f(x)$ to clearly sketch the following:

(i) $y = \frac{1}{f(x)}$

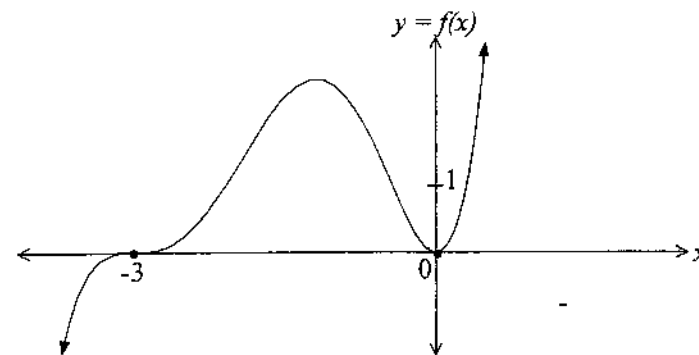


(ii) $y^2 = f(x)$



Please turn over for part (a)(iii).

(iii) $y = f'(x)$



Question 3 (b):

Possible polynomial equation for $y = f(x)$: _____

Suggested Solutions

Question 1

(a) (i) $\int \frac{x}{\sqrt{9-4x^2}} dx$

by inspection,

$$= -\frac{1}{4} \sqrt{9-4x^2} + C //$$

(or let $u = 9-4x^2$)

or $\frac{du}{dx} = -8x$
separating variables,
 $\therefore x dx = -\frac{1}{8} du$

$$\int \frac{x}{\sqrt{9-4x^2}} dx = -\frac{1}{8} \int \frac{du}{\sqrt{u}}$$

$$= -\frac{1}{8} \int u^{-1/2} du$$

$$= -\frac{1}{8} \left(\frac{u^{1/2}}{1/2} \right) + C$$

$$= -\frac{1}{4} \sqrt{u} + C$$

$$= -\frac{1}{4} \sqrt{9-4x^2} + C //$$

(ii) $\int \frac{x^2}{x+1} dx$

$$= \int \frac{(x-1)(x+1) + 1}{(x+1)} dx$$

$$= \int (x-1) + \frac{1}{x+1} dx$$

$$= \frac{x^2}{2} - x + \ln|x+1| + C$$

(or a substitution split)

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(iii) $\int_0^{\ln 2} x e^x dx$

let $u = x$ $dv = e^x$
 $du = 1$ $v = e^x$

$\therefore$ using $\int u dv = uv - \int v du$

$$= [x e^x]_0^{\ln 2} - \int_0^{\ln 2} e^x dx$$

$$= [x e^x - e^x]_0^{\ln 2}$$

$$= (\ln 2)(2) - e^{\ln 2} + e^0$$

$$= 2 \ln 2 - 2 + 1$$

$$= 2 \ln 2 - 1 //$$

(b) (i) $\frac{2}{(t+1)(t^2+1)} \equiv \frac{A}{t+1} + \frac{Bt+C}{t^2+1}$

$\Rightarrow 2 \equiv A(t^2+1) + (t+1)(Bt+C)$

by substitution:

$t = -1; \quad 2 = 2A$
 $\therefore A = 1$

$t = 0; \quad 2 = A + C$
 $2 = 1 + C$
 $\therefore C = 1$

$t = 1; \quad 2 = 2A + 2(B+C)$
 $2 = 2 + 2(B+1)$
 $\therefore B = -1$

$\therefore \frac{2}{(t+1)(t^2+1)} \equiv \frac{1}{t+1} - \frac{t+1}{t^2+1}$

(or use $t = \pm i$)

hence: $A = 1, B = -1, C = 1 //$

(ii) $\int_0^1 \frac{2}{(t+1)(t^2+1)} dt$

$$= \int_0^1 \left(\frac{1}{t+1} + \frac{1-t}{t^2+1} \right) dt$$

$$= \int_0^1 \frac{1}{t+1} + \frac{1}{t^2+1} - \frac{t}{t^2+1} dt$$

$$= \left[\ln|t+1| + \tan^{-1} t - \frac{1}{2} \ln(t^2+1) \right]_0^1$$

$$= \left[\tan^{-1} t + \ln \left| \frac{t+1}{\sqrt{t^2+1}} \right| \right]_0^1$$

$$= \tan^{-1} 1 + \ln \left| \frac{2}{\sqrt{2}} \right| - 0$$

$$= \frac{\pi}{4} + \ln \sqrt{2}$$

$$= \frac{\pi}{4} + \frac{1}{2} \ln 2 //$$

(iii) $t = \tan \frac{\pi}{4}$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{\pi}{4}$$

$$= \frac{1}{2} (\tan^2 \frac{\pi}{4} + 1)$$

$$\frac{dx}{dt} = \frac{1}{2} (t^2 + 1)$$

separating variables

$$\frac{2 dt}{t^2 + 1} = dx$$

$$\int_0^{\frac{\pi}{4}} \frac{2 dt}{1 + \tan^2 t} = \int_0^{\frac{\pi}{4}} dx$$

$$\text{also: when } x = 0, t = \frac{\pi}{4}$$

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}, \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\int_0^{\frac{\pi}{4}} \frac{2 \sin x}{1 + \sin^2 x} dx$$

$$= \int_0^1 \frac{2t}{1+t^2} \cdot \frac{1}{1+t^2} dt$$

$$= \int_0^1 \frac{2t}{(1+t^2)^2} dt$$

$$= \int_0^1 \frac{2}{(1+t^2)(t^2+1)} dt$$

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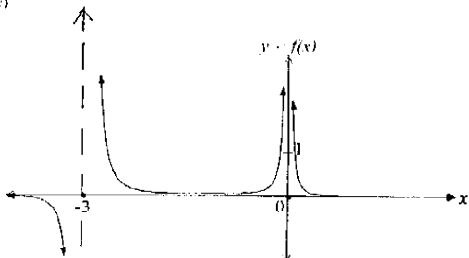
$$= \int_0^1 \frac{2}{(1+t^2)(t^2+1)} dt$$

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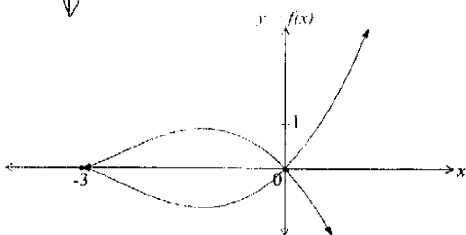
$$= \int_0^1 \frac{2}{(1+t^2)(t^2+1)} dt$$

Question 3(a)

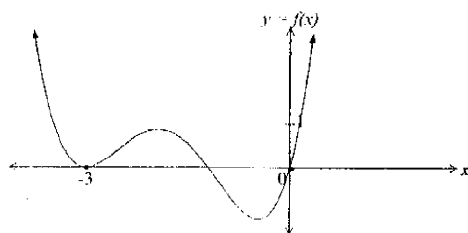
(i) $y = \frac{1}{f(x)}$



(ii) $y = f(x)$



(iii) $y = f'(x)$



(b) Possible equation: $y = ax^2(x+3)^3$ ($a \neq 0$)

Question 3

(c) (i) $1^2 + 3(1)^2 - 4$

$$= 1 + 3 - 4$$

$$= 4 - 4$$

$$= 0$$

$\therefore x = 1$ is a zero of

$$x^3 + 3x^2 - 4$$

(ii) other intercepts:

when $x = 0, y = -4 \Rightarrow (0, -4)$

also $x = 1$

N.B. $y = (x-1)(x+2)^2$

$\therefore x$ -intercepts are $x = 1$

$x = -2$

(double root)

$$\frac{dy}{dx} = 3x^2 + 6x$$

$$\text{IP's when } y' = 0$$

$$\text{or when } 3x(x+2) = 0$$

$$x = 0, x = -2$$

$$y'' = 6x + 6$$

$$y''(0) = 6 > 0 \text{ Rel. min.}$$

$$y''(-2) = -6 < 0 \text{ Rel. max.}$$

$$y''(x) = 6x + 6$$

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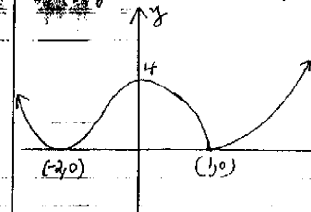
$$y''(x) = 6x + 6$$

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$$y''(x) = 6x + 6$$

(iii) $f(x) = |x^3 + 3x^2 - 4|$



(b) $y = \ln|x^3 + 3x^2 - 4|$

This is defined only with

$$y = |x^3 + 3x^2 - 4| > 0$$

However, we use the

graph we observe

$$y = \ln|x^3 + 3x^2 - 4|$$

is undefined when $x = -2, 1$

These are asymptotes.

Also, when $x = 0, y = \ln 4$

a max. T. as well.

$$\lim_{x \rightarrow \pm\infty} \ln|x^3 + 3x^2 - 4| = x$$

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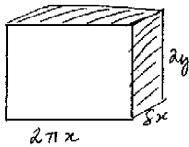
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$$\lim_{x \rightarrow \pm\infty} \ln|x^3 + 3x^2 - 4| = x$$

(vi)



$$V = 4\pi \int_0^a xy \, dx$$

$$= 4\pi \int_0^a x \left(\frac{b}{a} \sqrt{a^2 - x^2} \right) dx$$

(y > 0)

$$V = \frac{4\pi b}{a} \int_0^a x \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4\pi b}{a} \left[\frac{(a^2 - x^2)^{3/2}}{3} \right]_0^a$$

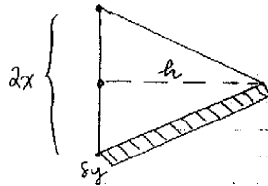
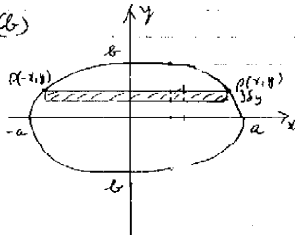
$$= \frac{4\pi b}{3a} [(a^2 - a^2)^{3/2} - (a^2)^{3/2}]$$

$$= \frac{4\pi b}{3a} [0 - a^3]$$

$$= \frac{4\pi b}{3a} [-a^3]$$

$$= \frac{4\pi b a^2}{3} [1 - 1] \text{ units}^3$$

(b)



Area of a typical horizontal slice

$$A(y) = \frac{1}{2} \cdot 2x \cdot h$$

$$A(y) = xh$$

Volume of a typical slice

$$\delta V = xh \delta y$$

$$\therefore \text{Total Volume} = \lim_{\delta y \rightarrow 0} \sum_{y=-b}^b xh \delta y$$

$$\therefore V = 2h \int_0^b x \, dy$$

$$\text{Since } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{a^2} = 1 - \frac{y^2}{b^2}$$

$$x^2 = a^2 \left(1 - \frac{y^2}{b^2} \right)$$

$$x = \frac{a}{b} \sqrt{b^2 - y^2}$$

$$V = \frac{2ha}{b} \int_0^b \sqrt{b^2 - y^2} \, dy$$

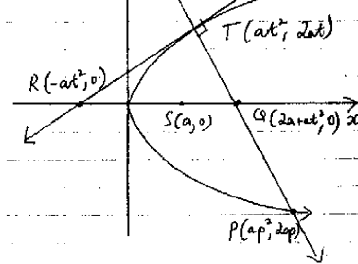
$$= \frac{2ha}{b} \left[\frac{1}{4} \pi b^2 \right]$$

$$= \frac{2ah \pi b^2}{4b}$$

$$V = \frac{\pi a b h}{2} \text{ units}$$

Question 5:

(a) (i)



$$(ii) \quad y = 4ax$$

$$2y \frac{dy}{dx} = 4a$$

$$\therefore \frac{dy}{dx} = \frac{2a}{y}$$

$$\text{when } y = 2at, \text{ at } T$$

$$\text{then } \frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$$

$$\text{Also, the slope of the normal at } T \text{ is } -t$$

$$\text{Equation of tangent:}$$

$$y - 2at = \frac{1}{t}(x - at^2)$$

$$x - ty = -at^2 \quad (1)$$

$$\text{Equation of normal:}$$

$$y - 2at = -t(x - at^2)$$

$$tx + y = 2at + at^3 \quad (2)$$

(iii) To find the length of RQ we need the coordinates of R & Q.

Equation of tangent meets x-axis at R when y = 0

$$\therefore \text{in (1)} \quad x = -at^2$$

$$\therefore R(-at^2, 0)$$

Equation of normal meets x-axis at Q when y = 0

$$\therefore \text{in (2)} \quad tx = 2at + at^3$$

$$(t \neq 0) \quad x = 2a + at^2$$

$$\therefore Q(2a + at^2, 0)$$

$$RQ = |-at^2| + |2a + at^2|$$

$$= at^2 + 2a + at^2$$

$$= 2at^2 + 2a$$

$$= 2a(t^2 + 1)$$

(iv) R will lie on the directrix if when x = -a (by definition y = 0)

$$\text{in (1)} \quad x - ty = -at^2$$

$$-a = -at^2$$

$$\therefore t^2 = 1$$

$$\Rightarrow t = \pm 1$$

(v) Since P(ap^2, 4ap^2) sat equation of normal at T

then in (2)

$$t(ap^2) + 4ap^2 = 2at + at^3$$

$$y: p^2t + 4p = 2t + t^3$$

$$y: 2(pt) = t(t^2 - p^2)$$

$$-2 = t(t + p) \quad (t \neq 0)$$

Question 6:

(a) (i)

$$v = 0, t = T, x = H$$

$$I_{mg} \uparrow \quad I_{k} \downarrow$$

$$x = 0, x = 0, v = 0$$

$$m \ddot{x} = \sum F_x$$

$$m \ddot{x} = -mg - mkv$$

$$\therefore \ddot{x} = -g - kv$$

$$\therefore \frac{dv}{dt} = -(g + kv)$$

(ii)

$$\frac{dt}{dv} = -\frac{1}{g + kv}$$

$$t = -\frac{1}{k} \ln |g + kv| + c_1$$

$$\text{when } t = 0, v = 0$$

$$\therefore 0 = -\frac{1}{k} \ln |g + k \cdot 0| + c_1$$

$$\therefore c_1 = \frac{1}{k} \ln |g|$$

$$\text{hence } t = \frac{1}{k} \ln |g + kv| - \frac{1}{k} \ln |g|$$

$$\therefore t = \frac{1}{k} \ln \left| \frac{g + kv}{g} \right|$$

(iii) when t = T, v = 0 (at max height)

$$\therefore T = \frac{1}{k} \ln \left(\frac{g + kv}{g} \right)$$

$$T = \frac{1}{k} \ln \left(1 + \frac{kv}{g} \right) \quad \text{from (ii)}$$

(iv) Considering $kt = \ln \left[\frac{g + kv}{g} \right]$

and solving for v

$$e^{kt} = \frac{g + kv}{g}$$

$$\therefore kv = (g + kv)e^{-kt} - g$$

$$v = \frac{1}{k} [(g + kv)e^{-kt} - g]$$

$$\text{let } v = \frac{dx}{dt}$$

$$\text{hence } \frac{dx}{dt} = \frac{1}{k} [(g + kv)e^{-kt} - g]$$

$$\text{hence } x = \frac{1}{k} \left[-\frac{1}{k} (g + kv)e^{-kt} - gt \right]$$

$$\text{when } t = 0, x = 0: C_2 = \frac{1}{k^2} (g + kv)$$

$$\text{hence } x = \frac{1}{k^2} (g + kv) [1 - e^{-kt}] - \frac{gt}{k}$$

$$\text{at } x = H, t = T$$

$$\therefore H = \frac{1}{k^2} (g + kv) [1 - e^{-kT}] - \frac{gT}{k}$$

$$\text{since } T = \frac{1}{k} \ln \left(1 + \frac{kv}{g} \right)$$

$$\text{then } kT = \ln \left(1 + \frac{kv}{g} \right)$$

$$e^{-kT} = \frac{g}{g + kv}$$

$$H = \frac{1}{k^2} (g + kv) \left[1 - e^{-\ln \left(\frac{g + kv}{g} \right)} \right] - \frac{gT}{k}$$

$$= \frac{1}{k^2} (g + kv) \left[1 - \frac{g}{g + kv} \right] - \frac{gT}{k}$$

$$= \frac{g + kv}{k^2} - \frac{g}{k^2} - \frac{gT}{k}$$

$$H = \frac{g + kv}{k^2} - \frac{g}{k^2} - \frac{gT}{k}$$

$$H = \frac{kv}{k^2} - \frac{gT}{k}$$

$$H = \frac{u - gT}{k}$$

$$\therefore H = \frac{1}{k} (u - gT)$$

$$(b) \quad I_n = \int_0^{\pi/2} \sin^n \theta \, d\theta$$

$$= \int_0^{\pi/2} \sin \theta \sin^{n-1} \theta \, d\theta$$

$$\text{let } u = \sin^{n-1} \theta \quad dv = \sin \theta$$

$$du = (n-1) \cos \theta \sin^{n-2} \theta \quad v = -\cos \theta$$

$$\therefore I_n = uv - \int v \, du$$

$$= [-\cos \theta \sin^{n-1} \theta]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \cos \theta \sin^{n-2} \theta \, d\theta$$

$$I_n = 0 + (n-1) \int_0^{\pi/2} \cos \theta \sin^{n-2} \theta \, d\theta$$

$$I_n = (n-1) \left[\int_0^{\pi/2} \sin^{n-2} \theta \, d\theta - \sin^n \theta \right]$$

$$I_n = (n-1) [I_{n-2} - I_n]$$

$$\therefore I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore n I_n = (n-1) I_{n-2}$$

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2} \quad (n \geq 2)$$

$$(i) \quad I_{10} = \frac{9}{10} I_8$$

$$= \frac{9}{10} \times \frac{7}{8} I_6$$

$$= \frac{9}{10} \times \frac{7}{8} \times \frac{5}{6} I_4$$

$$= \frac{9}{10} \times \frac{7}{8} \times \frac{5}{6} \times \frac{3}{4} I_2$$

$$= \frac{9}{10} \times \frac{7}{8} \times \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} I_0$$

$$I_0 = \int_0^{\pi/2} 1 \, d\theta$$

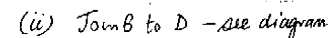
$$= \pi/2$$

$$I_{10} = \frac{9}{10} \times \frac{7}{8} \times \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$$

$$= \frac{945\pi}{7680}$$

$$= \frac{63\pi}{512}$$

(Q)



but $\angle TDB = \angle DCB$

(angle between tangent TD and chord BD is equal to the $\angle$ in the alt. segment)

hence: $LBDT = LBR T$

(iii) Town DR - not necessary though.

Since $L_{BDT} = L_{BRT}$ then

BRDT is a cyclic quadrilateral because the chord BT subtends equal angles at R & D - see diagram

(u) 2 Δ 's standing on the same chord are equal in this case they are

(iv) $BT = TD$ (two tangents drawn from the same ext. pt are =)

Since $\angle BRT$ is a cyclic quad. and since equal chords BT & TD subtend equal angles on the circumference then $\angle TRD = \angle BRT$ as well.

(v) $\angle DRT = \angle PDC$ (alt. $\angle$'s in parallel lines are equal)

$\angle BRT = \angle BCD$ from (ii)
 $\angle BRT = \angle CRT$ (from iv)
 $\therefore \angle RDC = \angle BCD = \angle CRD$
 $\therefore \angle RCD$ is isosceles.
 $(RC = RD \therefore)$

(vi) Join PC & QD
∴ PQDC is a cyclic quad rowel

$$LRPC = IPD - LPCD$$

(convert inches // lines)

$$\angle P Q D = 180 - \angle P C D$$

(Opp. Δ 's of a cyclic quad add to 180°)

$$\therefore \angle R P C = \angle P Q D$$

In $\triangle PRC$ & $\triangle QRD$
 $RC = RD$ (see (v))
 $\angle RPC = \angle QRD$ (from above)
 $\angle PRC = \angle QRD$ (alt $\angle$'s in $\parallel$ lines
 & $\perp$ secular Δ).

$\Delta PRC \equiv \Delta QPD$ (R.A.)
 $PR = RQ$ (Corresp. sides in
 $\equiv \Delta$ s are equal
 BC bisects PQ as required)

$$\begin{aligned} (b)(i) \quad \cos x &= \cos(-x) \\ &= \sin\left(\frac{\pi}{2} - (-x)\right) \\ &= \sin\left(\frac{\pi}{2} + x\right) \end{aligned}$$
$$\underline{\underline{\text{Q2:}}}$$

$$\sin\left(x + \frac{\pi}{2}\right) = \sin x \cos \frac{\pi}{2} + \cos x \sin \frac{\pi}{2}$$

$$= \cos x //$$

(ii) $y = 3 \sin x + 4 \cos x$
 $y = \sqrt{3^2 + 4^2} \sin(x + \alpha)$
 $y = 5 \sin(x + \alpha)$

By induction :

Prove the statement
 $\frac{d^n y}{dx^n} = 5 \sin(x + \alpha + \frac{n\pi}{2})$ is

for $n=1$
LHS = $\frac{dy}{dx} = 5 \cos(x + \pi)$

$$RHS = 5 \sin \left(x + \alpha + \frac{\pi}{2} \right)$$
$$= 5 \sin \left(\left(x + \frac{\pi}{2} \right) + \alpha \right)$$
$$= 5 \cos(x + \alpha)$$
$$= 5 \cos(x + \alpha) \quad \text{from (1)}$$

just replace x with $(x + \alpha)$

Assume the statement is true for $n=k$ $\square 1 \leq k \leq n$
 $(k, n) \in \mathbb{Z}$

RTP the statement is true
for $n = k+1$.

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Closin

$$\frac{dy}{dx} = 5 \sin \left[(x + \alpha) + \frac{\pi}{2} k \right]$$

ATP

$$\frac{d^{\text{ker}} y}{d^{\text{ker}} x} = 5 \sin \left[(x^{\text{ker}}) + \left(\frac{\text{ker}}{2} \right) \frac{\pi}{2} \right]$$

From (2)

$$\text{LHS} = \frac{d}{dx} \ln y$$
$$= \frac{d}{dx} \left(\frac{dy}{dx} \right)$$
$$= \frac{d}{dx} \left[5 \sin \left[(x + \alpha) + \frac{\pi}{2} \right] \right]$$
$$= 5 \cos \left[\omega t + \frac{k\pi}{2} \right] \quad \text{from}$$

$$= 5 \sin \left[\frac{\pi}{2} + k\pi \right]$$

[just replace $x \rightarrow x$
in (i)]

$$= 5 \sin \left[\frac{\pi}{a} (b + 1) + (x + a) \right]$$

The statement is true for whenever it is true for n it is also true for $n+1$ and for all positive integer values for $n > 1$.